REP-113 · Do standing orders pay, with the stake and the cost of the fruit?

Game	Fruit Market 1.20.0 live, rules version 18 (the working branch has the same rules)
Report date	2026-11-12 (Asia/Kuala_Lumpur)
Type	Analysis for a decision
Request	Owner, 2026-11-12: "analyse if standing order makes sense with the cost involved"
Approval	Nothing is changed by this report. Options in section 5 are for the owner to choose.

1. Short answer

Yes for a careful bidder, but only just. Winning a standing order at a sensible bid roughly doubles what each piece earns, yet over a whole game it adds only about $1 to $1.60 to a stall's cash (about 1 to 2 per cent), because the order uses the same day's buying, shelf and cash that would otherwise sell in the rush. Reckless bids lose a lot: the stake does its job. Mango orders are the weak spot: about half find no bidder, and half of those taken fail.

2. What one order costs and pays (the card numbers)

Order	Highest bid	Stake	Pieces at 2–3 / 4 / 5–6 stalls	Cost to buy a piece	Profit a piece at the highest bid	The same piece sold in the rush (average sale price less cost, 4 stalls)	Cash needed the day you win, 4 stalls
Blueberry	$9	$6	3 / 4 / 5	$2 Fresh, $1 Okay	$7	$2.42	$6, the pieces later (they last 3 days, so buy them on the last two days)
Watermelon	$12	$8	3 / 4 / 5	$3	$9	$2.81	$8, the pieces over three buying turns
Apple	$11	$7	3 / 4 / 5	$3	$8	$2.36	$7, the pieces over three buying turns
Mango	$18	$12	3 / 4 / 5	$4 Green	$14	$6.51	$28: the stake and four Green mangoes that same day, so they ripen in time

An Order Book halves the stake and pays $2 more a piece.

3. What happens in play (computer games, rules 18 as built)

1,500 games each at 3, 4 and 6 stalls with Normal computers, and 1,000 at 4 stalls with Hard computers, 10 days.

4 stalls, Normal	Blueberry	Watermelon	Apple	Mango
Nobody bid	5%	2%	4%	56%
Bidders an order	3.2	2.9	2.9	0.9
Winning bid (share of the highest bid)	$6.29 (70%)	$7.82 (65%)	$7.44 (68%)	$13.29 (74%)
Paid a piece, with the Order Book's $2	$8.05	$9.34	$8.92	$14.86
Delivered in full	76%	83%	80%	51%
Stake lost for each order held	$1.31	$1.29	$1.23	$5.01
The holder's margin on the order (paid less cost, less the stake lost)	$20.38	$22.19	$20.21	$27.18

At 3 stalls mango orders find no bidder 44% of the time; at 6 stalls 49%, and only 39% of those taken are delivered in full. Hard computers bid lower (57 to 68% of the highest bid) and deliver a little more often.

The stall that holds an order ends its three days $7.53 (3 stalls), $11.71 (4) and $15.86 (6) ahead of the average rival. That flatters orders: the holder is the stall that was best placed to bid lowest.

4. The fair test: the same seat, made to bid one way (600 paired games a row)

Seat's rule	Cash against bidding as the computer does, 3 stalls	4 stalls
Bids as the computer does (weighs whether it can deliver)	0	0
Always passes	−$1.57 (SE 0.64)	−$0.88 (SE 0.58)
Always bids the highest bid	−$0.93 (SE 0.74)	−$0.80 (SE 0.73)
Always bids 60% of the highest bid	−$6.58	−$3.84
Always bids half	−$10.20	−$7.95
Always bids $1	−$28.92	−$31.78

So: a careful bid is worth about $1 to $1.60 a game over never bidding; bidding the highest bid wins too seldom to matter; low bids lose stakes (about $4 to $7 a game) and short deliveries.

5. Why the margin is large but the gain is small

- **The order replaces rush sales, it does not add to them.** A stall buys at most 4 pieces a day (5 with a Handcart) and holds 8. Pieces bought for the order are pieces not bought for the row.
- **The cash is tied up on the morning it is won**, when a stall usually has $15 to $25: the stake comes out before buying.
- **Every stall wants the same lorry.** When the lorry brings few of that fruit, the holder can come up short and lose the stake.

- **Mango** needs the stake and the Green mangoes on the same day ($24 to $32), which most stalls do not have on Day 3, and Green mangoes are scarce.

6. Options for the owner

1. **Leave it as it is.** It is a small plus for planning ahead and the stake stops wild bids. Mango orders stay mostly unbid (DEC-101 kept them for now).
2. **Show the sums on the bid sheet (no rule change).** For example: "4 watermelons cost about $12 at the market today. At $10 a piece you clear about $28 if you deliver all 4." This helps a person bid well without changing the game. Recommended in any case.
3. **Make mango orders workable (a rules change, rules version 19).** For example: a mango order is due in 3 days instead of 2, so the mangoes can be bought the day after winning; or the mango stake is $8. Needs a balance run first.
4. **Make orders worth more (a rules change).** For example: the order's holder buys first on the due day, so it is less often short; or a full delivery pays a $3 bonus. Needs a balance run first.

7. How the numbers were made

`docs/studio/review-17/balance/scripts18: sim.ts` batches (seeds 1 up, 10 days, all events and upgrades), `analyze.ts --sections 2,7`, `focal.ts --test orders` (one seat follows a rule, the others play as usual; same seeds for every row, so the differences are paired). Computer players only; people may bid differently. Evidence in the session's scratchpad folder `so/`.